Hospital Ownership Type Correlated With Investments in Nursing Services

Evidence from Illinois

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Background: The association between hospital ownership type and patient care quality has garnered increased public attention.

Objectives: To describe differences in investments in nursing services, care quality and safety, and nurse job outcomes among for-profit and not-for-profit hospitals in Illinois.

Research Design: Cross-sectional, descriptive case study of hospital nursing services, patient care quality and safety outcomes, and nurse job outcomes in 113 hospitals in Illinois in 2021. Three datasets were linked through a common identifier: RN4CAST-IL nurse survey, National Academy for State Health Policy (NASHP) cost data, American Hospital Association (AHA) Annual Hospital Survey.

Measures: Nursing services included patient-to-nurse staffing ratios and the quality of the nurse work environment. Nurse job outcomes included burnout, job dissatisfaction, and intent to leave.

Results: For-profits had statistically significantly lower investments in nursing services, worse nurse job outcomes, and poorer patient care quality and safety outcomes relative to not-for-profit hospitals, despite having no statistically significant differences in operating margins. For-profit hospitals had poorer nurse work environments (mean score 2.5 vs. 2.9, $P<.001$) and worse staffing ratios (6.8 vs. 4.7 patients per nurse, $P<0.01$), higher percentages of nurses intending to leave their employer (36.2% vs. 22.8%, $P<.01$), higher job dissatisfaction (37.9% vs. 23.4%, $P<.01$), higher burnout (63.8% vs. 46.2%, $P<.05$), and worse quality and safety (e.g., poor patient safety grades 53.6% vs. 33.8%, $P<.01$).

Conclusions: Investments in nursing services as well as quality and safety of patient care in for-profit hospitals are worse than in not-for-profit hospitals, despite having no significant differences in operating margins.

Key Words: hospital, Illinois, nursing, ownership, profit status

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Current evidence shows that on average, for-profit hospitals have higher rates of mortality[1,2] and rehospitalizations,[3] poorer access,[4] worse quality, and lower cost-efficiency.[5] It is unknown what explains the poorer outcomes among for-profit hospitals. In this study, we test a novel hypothesis that one explanation is that for-profit hospitals invest in their nursing services to a lesser degree than not-for-profit hospitals.

Decades of evidence have shown that hospitals with more favorable nursing services (i.e., patient-to-nurse staffing, nurse work environments) have better outcomes for patients[6–9] and nurses (e.g., lower burnout).[6,10] In this descriptive case study, we explore whether nursing services and quality measures differ between for-profit and not-for-profit hospitals. In addition, we evaluate differences in operating margins, as they could explain potential differences in nursing services.

METHODS

Study Design and Sample

This exploratory cross-sectional study used 3 datasets linked through a common hospital identifier: 2021 RN4CAST-IL survey of registered nurses (RNs), 2020 American Hospital Association (AHA) Annual Hospital Survey, 2021 National Academy for State Health Policy (NASHP) cost data. The RN4CAST-IL survey provided data from RNs licensed to work in Illinois. Nurses received an electronic survey through email from April to June 2021 asking about their hospital's attributes. Survey methods have been described in detail elsewhere.[11]

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The authors declare no conflict of interest.

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AHA data provided information on hospital characteristics: ownership type, size, teaching status, technology capabilities. NASHP provided financial data from hospital accounting records to calculate hospital operating margins.

The final analytic sample included 113 non-federal adult acute care general hospitals in Illinois with an average of 41 nurse respondents per hospital (range: 5–330).

Measures

Ownership Type

For-profit hospitals are defined as investor-owned. Not-for-profit hospitals are defined as non-government institutions that are either church-operated or other non-investor-owned.

Operating Margin

Operating margin, derived from NASHP cost data,[12] is an overall measure of hospital financial performance that includes labor costs (e.g., nursing costs) and income generated from providing care. For each hospital, operating margin was calculated by dividing the difference between net patient revenue and operating expense by net patient revenue and multiplied by 100 (**Equation 1**); where: net patient revenue is defined as gross patient charges, less contractual discounts, bad debt and charity care allowances, and other deductions agreed to by the hospital.

$$\textit{Operating margin} = \frac{(\textit{net patient revenue} - \textit{operating expense})}{\textit{net patient revenue}} * 100 \quad (1)$$

Patient Quality and Safety

Validated indicators of patient safety and care quality were derived from the RN4CAST-IL survey and aggregated to the hospital-level. Nurse respondents provided overall grades for their hospital on patient safety and infection prevention (A = excellent, B = good, C = acceptable, D = poor, F = failing). We report the hospital percentage of nurses who gave unfavorable grades (C, D, or F) on patient safety and infection prevention. Nurses also rated the quality of care in their hospital on a 4-point Likert scale ranging from "poor" to "excellent". Nurses were asked whether they would recommend their hospital to friends/family needing care on a 4-point Likert scale. We computed the hospital percentage of nurses who did not say they would 'definitely' recommend their hospital.

We report percentages of nurses who 'strongly' or 'somewhat' agreed with the 7 items from the AHRQ Patient Safety Culture tool[13] and the percentage of nurses reporting their work is frequently interrupted/delayed by operational failures.[14] Nurses reported on a 4-point Likert scale how confident they were that their management could act to resolve problems they identified in patient care, for which we reported the percentage of nurses who were "somewhat" or "not at all confident."

Nurse Job Outcomes

Nurse job outcomes included the hospital percentage of nurses with high burnout, job dissatisfaction, and intent to leave. High burnout was measured using the Maslach Burnout Inventory emotional exhaustion subscale, defined as a score greater than or equal to 27[15]. Job dissatisfaction was derived through a single-item measure: "Overall, how satisfied are you with your job?" Intent to leave was measured through a binary single-item question: "Do you plan to be with your current employer 1 year from now?" We report hospital average percentages of nurses experiencing each outcome.

Nursing Services

Measures of hospital nursing services included patient-to-nurse staffing ratios and ratings of the nurse work environment, derived from the RN4CAST-IL. Patient-to-nurse staffing was measured by the total number of patients that nurses reported were on their unit during their last shift, divided by the number of nurses on the unit. Staffing was aggregated to the hospital-level by computing the mean response among nurses in the same hospital.

The nurse work environment was measured through the abbreviated Practice Environment Scale of the Nursing Work Index-5 (PES-5)[16]. Nurses indicated their level of agreement on organizational attributes present in their jobs, including (1) administration that listens and responds to nurses concerns, (2) a nurse manager who is a good manager and leader, (3) a lot of team work between nurses and physicians, (4) enough staff to get the work done, and (5) a clear philosophy of nursing that pervades the patient care environment. Nurse responses were averaged to create a hospital-level, continuous measure of the work environment.

Additional Hospital Characteristics

Hospital size was defined by the number of beds, categorized as small (≤ 100 beds), medium (101–250 beds) and large (> 250 beds). Teaching status was based on the ratio of medical residents to fellows per bed (non-teaching had no medical residents/fellows; minor teaching $< 1{:}4$; major teaching $\geq 1{:}4$). High-technology hospitals had capacity to perform open-heart surgery or major organ transplantation; non-high-technology hospitals did not.

Data Analysis

Descriptive statistics compared hospital mean staffing, nurse work environments, patient care outcomes, nurse job outcomes, operating margins, and hospital characteristics among for-profit and not-for-profit hospitals. Fisher exact tests were used for categorical measures and student *t* tests for continuous measures. Scatterplots presented correlations between hospitals' nurse work environment (Fig. 1), and staffing (Fig. 2) and operating margins. Data were analyzed using Stata 17.16.

RESULTS

Among the 113 study hospitals in Illinois, 8 hospitals were for-profit and 105 hospitals were not-for-profit

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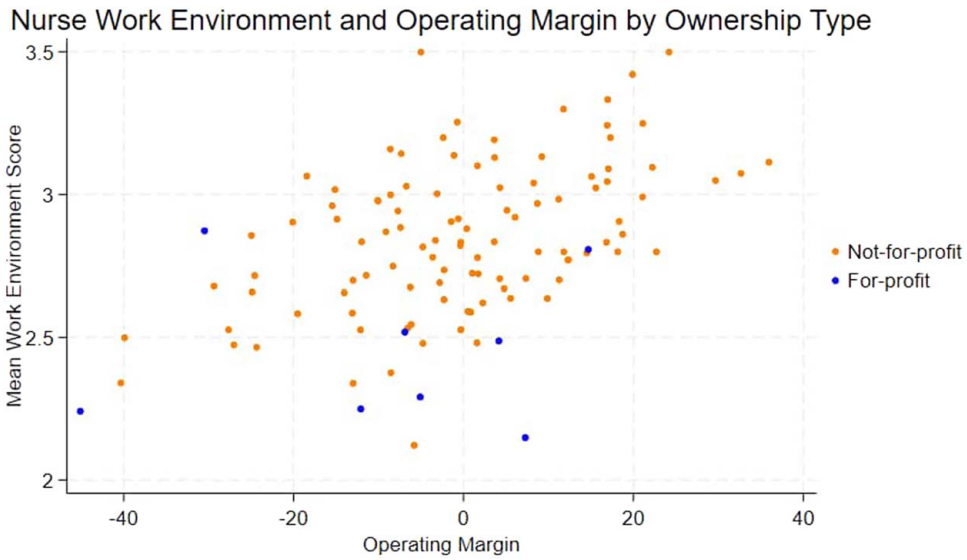

FIGURE 1. Relationship between hospital operating margins and nurse work environment score among for-profit and not-for-profit hospitals. Correlation of operating margins and nurse work environment score in not-for-profits ($r = 0.4857$, $P < .001$) and for-profits ($r = 0.0705$, $P = 0.87$). full color online

(Table 1). Given the small number of for-profit hospitals, the findings are primarily descriptive.

We found clinically (and statistically significant) differences in nursing services between for-profit and not-for-profit hospitals (Table 1). On average, nurses in for-profit hospitals reported caring for 6.8 patients at a time, relative to nurses in not-for-profit hospitals who cared for 4.7 patients a time ($P < 0.001$). For-profit hospitals had lower average scores on the nurse work environment scale, (2.5 vs. 2.9, $P < 0.001$) indicating nurses in for-profit hospitals on average reported less favorably about their work environments.

There were no statistically significant differences in average operating margins between for-profit and not-for-profit hospitals, although the average operating margin in for-profits was −9.2% compared with −0.1% in not-for-profits (Table 2, $P = 0.11$). There were no statistically significant differences in hospital characteristics (i.e., size, teaching status, technology capabilities) by ownership type.

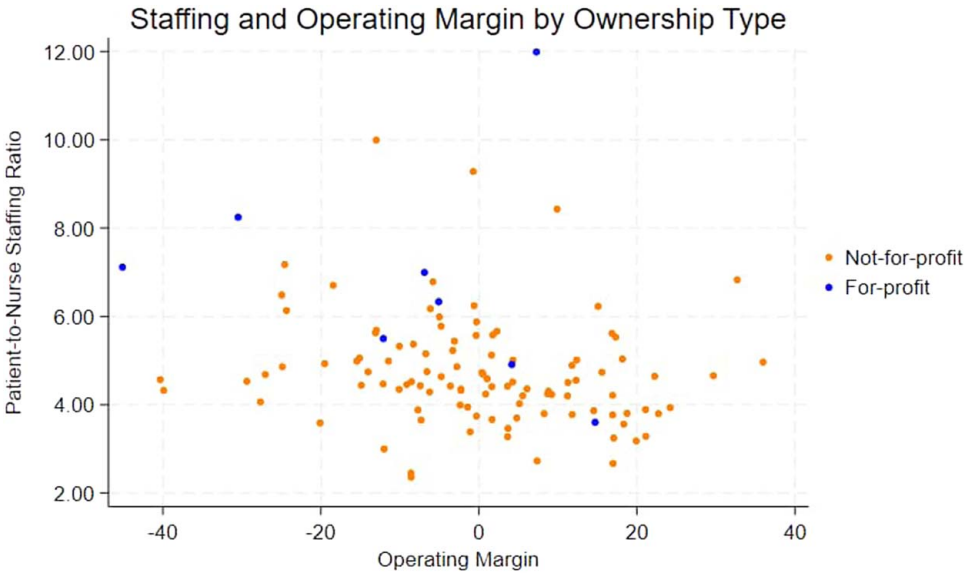

FIGURE 2. Relationship between hospital operating margins and patient-to-nurse staffing ratios among for-profit and not-for-profit hospitals. Correlation of operating margins and patient-to-nurse staffing in not-for-profits ($r = -0.1732$; $P = 0.08$) and for-profits ($r = -0.1575$; $P = 0.71$). full color online

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TABLE 1. Nursing Services Among For-Profit and Not-For-Profit Hospitals

Characteristic	All (n = 113)	For-profit (n = 8)	Not-for-profit (n = 105)	*P*
Patient-to-nurse staffing, mean (SD)	4.9 (1.5)	6.8 (2.5)	4.7 (1.2)	< 0.001
Nurse work environment, mean (SD)	2.8 (0.3)	2.5 (0.3)	2.9 (0.3)	< 0.001

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Among the 8 for-profit hospitals, the correlation between the nurse work environment score and operating margin was weak and not statistically significant (Fig. 1, $r = 0.07$, $P = 0.87$). Among the 105 not-for-profit hospitals, the correlation was considerably stronger and statistically significant ($r = 0.49$, $P < 0.001$) such that more favorable nurse work environment scores were positively associated with higher operating margins. Correlations between nurse staffing and operating margins were weak and not statistically significant for both not-for-profit (Fig. 2, $r = -0.17$; $P = 0.08$) and for-profit hospitals ($r = -0.16$; $P = 0.71$).

On average, worse patient care quality and safety were observed in for-profit hospitals compared with not-for-profit hospitals (Table 3). Statistically and clinically significant higher percentages of nurses reported unfavorable patient safety grades (53.6% vs. 33.8%, $P < .01$) and unfavorable infection prevention grades in for-profits (49.2% vs. 23.8%, $P < .001$). Higher percentages of nurses would not 'definitely' recommend their hospital to family or friends in for-profit compared with not-for-profit hospitals (60.3% vs. 23.3%, $P < .001$). For-profit hospitals also performed worse on measures of culture of patient safety and operational failures. For example, higher percentages of nurses in for-profit hospitals reported that mistakes were held against them (45.4% vs. 33.6%, $P < .05$), important information is lost during shift changes (47.9% vs. 36.1%, $P < .05$), feedback about changes were not put into place based on event reports (69.3% vs. 39.4%, $P < .001$), that the actions of management do not show that patient safety is a top priority (65.4% vs. 42.8%, $P < .001$), and that discussions were not in place to discuss ways to prevent errors from happening (43.7% vs. 21.2%, $P < .001$).

TABLE 2. Operating Margins and Hospital Characteristics Among For-Profit and Not-For-Profit Hospitals

Hospital characteristics	All (n = 113)	For-profit (n = 8)	Not-for-profit (n = 105)	*P*
Operating margin, % (SD)	−0.7 (15.4)	−9.2 (20.0)	−0.1 (14.9)	0.11
Bed size, n (%)				0.27
Small	26 (23.0)	3 (37.5)	23 (21.9)	
Medium	43 (38.1)	4 (50.0)	39 (37.1)	
Large	44 (38.9)	1 (12.5)	43 (41.0)	
Teaching status, n (%)				0.24
Non-teaching	68 (60.2)	3 (37.5)	65 (61.9)	
Minor teaching	32 (28.3)	4 (50.0)	28 (26.7)	
Major teaching	13 (11.5)	1 (12.5)	12 (11.4)	
Technology status, n (%)				0.30
Not high-tech	63 (55.8)	6 (75.0)	57 (54.3)	
High-tech	50 (44.2)	2 (25.0)	48 (45.7)	

For-profit hospitals also had higher percentages of nurses reporting low confidence that management would resolve problems in clinical care that nurses identified (79.7% vs. 66.4%, $P < .05$). Nurses in these hospitals also reported having their work frequently interrupted or delayed because of insufficient staff (68.2% vs. 47.9%, $P < .01$). Not-for-profits performed worse on only one indicator that was marginally significant—that nurses do not feel free to question authority (48.4% vs. 33.6%, $P < .05$).

Nurse job outcomes were worse for nurses employed in for-profit hospitals, including higher percentages of nurses who were intending to leave their employer within a year (36.2% vs. 22.8%, $P < .01$), having high burnout (63.8% vs. 46.2%, $P < .05$), and reporting job dissatisfaction (37.9% vs. 23.4%, $P < .01$).

DISCUSSION

We found that for-profit hospitals invest in their nursing services to a lesser degree than not-for-profit hospitals and have poorer patient safety and quality of care and worse nurse job outcomes. The lower investment in nursing services among for-profits is not associated with differences in operating margins or other hospital characteristics like size, for which we found no statistical differences. The study was limited to a single state (Illinois) with only 8 for-profit hospitals. Nevertheless, the findings add a novel contribution to understanding why for-profit hospitals, on average, confer poorer care quality.

It's notable that we did not observe statistically significant differences in operating margins between for-profit and not-for-profit hospitals but found large and statistically significant differences in the extent to which for-profit and not-for-profit hospitals, on average, invested in their nursing services. Nurses in for-profit hospitals, on average, were assigned more patients per nurse (nearly 2 patients per nurse more) and they rated their work environments less favorably. The finding suggests the possibility that for-profit hospitals spend less on nursing services—namely, hiring enough bedside nurses and enhancing the work environment—relative to not-for-profit hospitals.

We found weak correlations between operating margins and nurse staffing, which runs counter to commonly cited concerns that mandatory hospital minimum nurse staffing requirements would be financially untenable.[17] Irrespective of hospital ownership type, we observed hospitals with high and low operating margins have high and low staffing, which suggests that strong investments in nurse staffing are possible irrespective of operating margins.

Higher operating margins were significantly positively correlated with better nurse work environments

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TABLE 3. Patient Safety, Quality of Care, and Nurse Job Outcomes Among For-Profit and Not-For-Profit Hospitals

Characteristic, % (SD)	All (n = 113)	For-profit (n = 8)	Not-for-profit (n = 105)	*P*
Patient safety and quality of care				
Unfavorable patient safety grade	35.2 (18.8)	53.6 (15.5)	33.8 (18.3)	< 0.01
Unfavorable infection prevention grade	25.6 (18.1)	49.2 (13.4)	23.8 (17.2)	< 0.001
Would not "definitely" recommend hospital to friends/family	25.9 (22.1)	60.3 (21.3)	23.3 (19.9)	< 0.001
Poor/Fair quality of care	19.1 (22.2)	26.7 (22.2)	18.5 (15.9)	0.18
Less than confident that management can resolve clinical care problems that nurses identify	67.3 (15.6)	79.7 (17.6)	66.4 (15.1)	< 0.05
Frequently interrupted by insufficient staff	49.3 (20.7)	68.2 (20.6)	47.9 (20.0)	< 0.01
Culture of patient safety				
Staff feel mistakes are held against them	34.4 (16.1)	45.4 (19.2)	33.6 (15.6)	< 0.05
Important information is lost during shift changes	36.9 (14.6)	47.9 (21.5)	36.1 (13.8)	< 0.05
Things fall between the cracks	42.9 (16.2)	44.9 (20.0)	42.8 (15.9)	0.73
Staff do not feel free to question authority	47.4 (16.3)	33.6 (18.0)	48.4 (15.7)	< 0.05
Staff don't discuss ways to prevent errors from happening again	22.8 (14.9)	43.7 (13.4)	21.2 (13.9)	< 0.001
Feedback about changes are not put into place based on event reports	41.5 (17.4)	69.3 (20.3)	39.4 (15.3)	< 0.001
Patient safety is not a top priority of management	44.4 (18.3)	65.4 (15.6)	42.8 (17.5)	< 0.001
Nurse job outcomes				
High burnout	47.5 (19.2)	63.8 (28.6)	46.2 (17.9)	< 0.05
Intent to leave	23.8 (14.2)	36.2 (10.0)	22.8 (14.0)	< 0.01
Job dissatisfaction	24.5 (13.2)	37.9 (14.9)	23.4 (12.5)	< 0.01

among not-for-profit hospitals, but not significantly correlated among for-profit hospitals; which suggests not-for-profit hospitals may invest marginal operating revenue on improvements to the work environment, whereas for-profit hospitals do not. Our findings support existing policy recommendations for hospitals to adequately staff nurses in hospitals, and to invest in nurses' work environments[8,16] through nurse autonomy and involvement in hospital affairs, interprofessional collaboration, responsive hospital management, and foundations for high-quality patient care. The value of such investments is evident in California where safe hospital nurse staffing legislation[18–20] has resulted in favorable patient care and nurse job outcomes.

The study has notable limitations, including that the sample has only 8 for-profit hospitals; however, even with limited statistical power we detected clinically meaningful and statistically significant differences in nursing services. Other limitations include the cross-sectional design, which limits our ability for causal inference, and correlation and bivariate comparisons that lack adjustment for potential confounders. Future studies with a larger sample size of for-profit hospitals should consider more robust analyses to adjust for other factors that may influence operating margins, patient outcomes, and nurse outcomes. For example, we did not account for pandemic-related factors that may have impacted operating margins and nursing services.

CONCLUSIONS

This descriptive case study is the first to identify large differences in the extent to which for-profit and not-for-profit hospitals invest in nursing services. We find that for-profit hospitals have significantly worse patient-to-nurse staffing and poorer work environments, which is a likely explanation for the poorer patient safety and quality of care and worse nurse job outcomes also observed among for-profit hospitals. The lower investment in nursing services among for-profits is not explained by differences in operating margins, or other hospital characteristics like size, which we found did not differ significantly by hospital ownership.

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